Section 1: Principles of chemistry

a) States of matter

b) Atoms

c) Atomic structure

d) Relative molecular and formula masses

e) Chemical formulae and chemical equations

f) Ionic compounds

g) Covalent substances

h) Metallic crystals

i) Electrolysis

a) States of matter

Students will be assessed on their ability to:

1.1 understand the arrangement, movement and energy of the particles in each of the three states of matter: solid, liquid and gas

1.2 describe how the interconversion of solids, liquids and gases are achieved and recall the names used for these interconversions

1.3 describe the changes in arrangement, movement and energy of particles during these interconversions

b) Atoms

Students will be assessed on their ability to:

1.4 describe simple experiments leading to the idea of the small size of particles and their movement including:

- i dilution of coloured solutions
- ii diffusion experiments

1.5 understand the terms atom and molecule

1.6 understand the differences between elements, compounds and mixtures

1.7 describe techniques for the separation of mixtures, including simple distillation, fractional distillation, filtration, crystallisation and paper chromatography

c) Atomic structure

Students will be assessed on their ability to:

1.8 recall that atoms consist of a central nucleus, composed of protons and neutrons, surrounded by electrons, orbiting in shells

1.9 recall the relative mass and relative charge of a proton, neutron and electron

1.10 understand the terms atomic number, mass number, isotopes and relative atomic mass (A^r)

1.11 calculate the relative atomic mass of an element from the relative abundances of its isotopes

1.12 understand that the Periodic Table is an arrangement of elements in order of atomic number

1.13 deduce the electronic configurations of the first twenty elements from their positions in the Periodic Table

1.14 deduce the number of outer electrons in a main group element from its position in the Periodic Table

d) Relative formula masses and molar volumes of gases

Students will be assessed on their ability to:

1.15 calculate relative formula masses (M^r) from relative atomic masses (A^r)

1.16 understand the use of the term mole to represent the amount of substance

1.17 understand the term mole as the Avogadro number of particles (atoms, molecules, formulae, ions or electrons) in a substance

1.18 carry out mole calculations using relative atomic mass (A^r) and relative formula mass (M^r)